

		Parent's signature
Name	Index Number	CTG

YISHUN JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2018

BIOLOGY

8876/02

HIGHER 1

29 AUGUST 2018

Paper 2 Structured and Free-response Questions

Wed 1400 - 1600

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.



READ THESE INSTRUCTIONS FIRST

Write your name and CTG in the spaces at the top of this page and on all separate answer paper used.

Write in dark blue or black pen only.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question paper.

Section B

Write your answers in the spaces provided on the Question Paper.

At the end of the examination, fasten all your work securely together.

This question paper consists of **20** printed pages.

FOR EXAMINER'S USE	
Paper 1	/ 30
Paper 2	
Section A	
Q1	/14
Q2	/10
Q3	/ 6
Q4	/ 5
Q5	/ 10
Section B	
Q6 OR 7	/ 15
Total	/ 60
Overall	/ 90

Section A

Answer **all** questions in this section.

- 1** In an experiment to study the effect of heat treatment on the digestibility of protein substrate and the effect of raw bean extract on protease activity, various reaction mixtures were prepared and were incubated for 30 minutes.

The protein concentration of each reaction mixture at the beginning and at the end of the incubation period was determined by the colorimetric method which measures colour intensity of these reaction mixtures. The results were shown in Table 1.1.

Table 1.1

Incubation period /min	Colour intensity of the reaction mixture / arbitrary unit		
	Tube A	Tube B	Tube C
	Protease + heated protein substrate	Protease + unheated protein substrate	Protease + unheated protein substrate + raw bean extract
0	10	10	10
30	4	7	9

Fig. 1.1 shows a standard graph obtained by using colorimetric method for determining concentration of protein solutions.

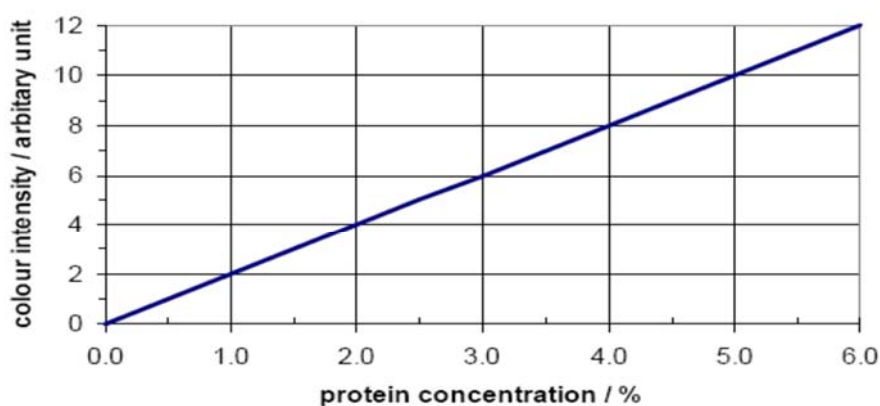


Fig. 1.1

- (a) With reference to Table 1.1 and Fig 1.1,

- (i) Determine the protein concentration of the three tubes at time 0 min.

..... [1]

- (ii) Explain which protein substrate is more digestible by protease.

.....

.....

.....

..... [2]

- (b) Based on your knowledge of protein structure, explain the effect of heat treatment on the digestibility of this protein substrate.

.....

.....

.....

.....

.....

..... [3]

- (c) Suggest the effect of raw bean extract on protease activity.

..... [1]

- (d) Dextrin is a generic term applied to a variety of products obtained by heating starch in the presence of small amounts of moisture and an acid. Dextrins can be made from starch and is a polysaccharide similar to amylopectin.

Describe the differences in molecular structure between cellulose and dextrins.

.....

.....

.....

..... [2]

Fig. 1.2 shows the structure of a triglyceride. Fig. 1.3 shows the structure of a glycolipid.

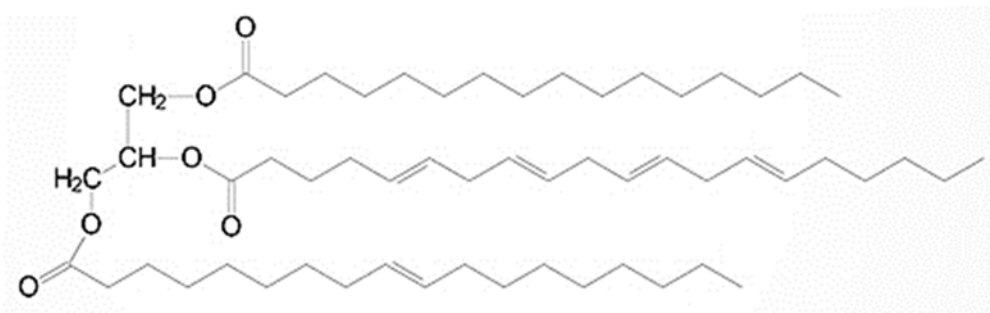


Fig. 1.2

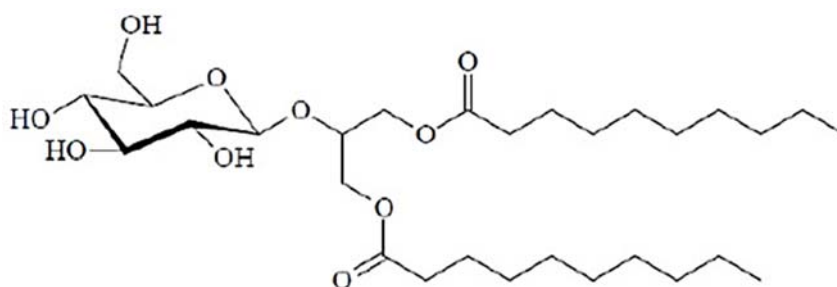


Fig. 1.3

- (e) Explain the difference in solubility between triglycerides and the products of their hydrolysis.

.....

 [3]

- (f) Describe the roles of triglycerides and glycolipids in relation to the differences in their molecular structures.

.....

 [2]

[Total: 14]

- 2 Fig. 2.1 shows two pairs of homologous chromosome carrying alleles **F/f** and **G/g** respectively.

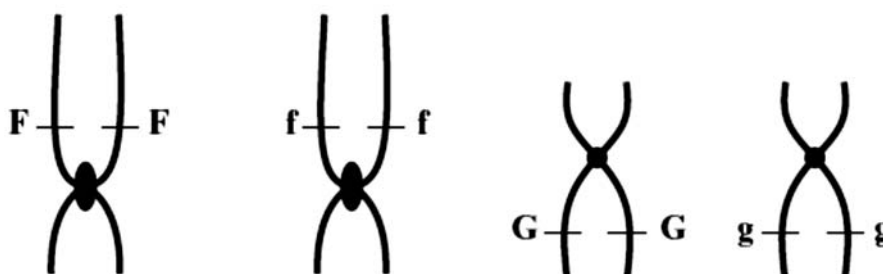


Fig. 2.1

- (a) State the genotype of this cell.

..... [1]

- (b) Explain why there are two copies of each allele for each gene present on the chromosomes in Fig. 2.1, before the start of nuclear division.

.....

 [2]

Fig. 2.2 shows a molecule of tRNA and the enzyme that attaches the correct amino acid to it.

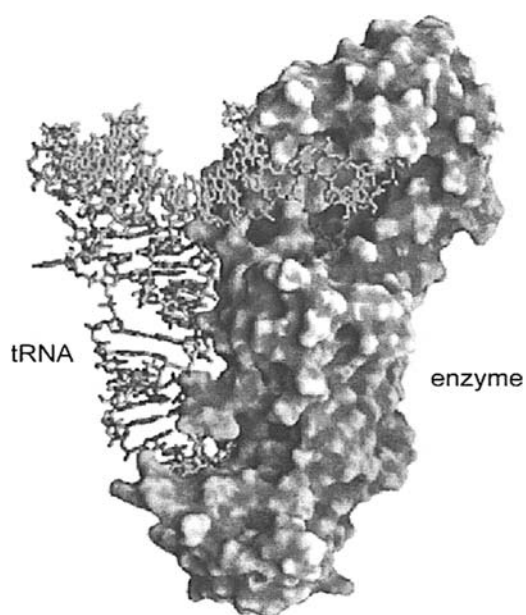


Fig. 2.2

- (c) Describe **two** structural features which adapt tRNA to its role in translation.

.....

.....

..... [2]

- (d) Explain how the enzyme in Fig. 2.2 is suited to its function.

.....

.....

.....

..... [3]

- (e) Fig. 2.3 below shows the base sequence on a length of mRNA and the amino acids represented by the respective codons.

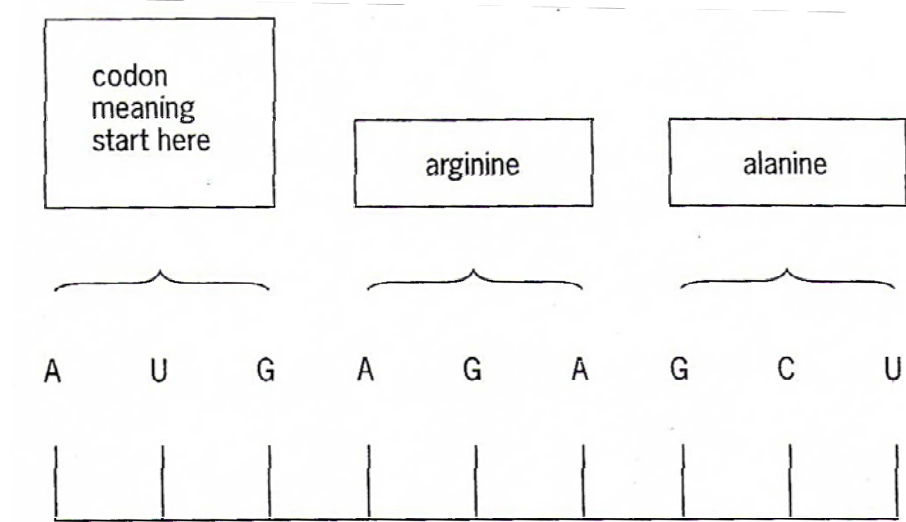


Fig. 2.3

The codon AGA codes for the amino acid arginine. However, arginine can also be represented by the codon AGG.

- (i) Explain why the substitution of A by G in this case does not affect the amino acid represented.

.....
..... [1]

- (ii) The codons in the mRNA sequence do not overlap. Suggest one advantage of this.

.....
..... [1]

[Total: 10]

- 3** An inbred variety of maize, **A**, with finely striped leaves was found to be resistant to the fungus that causes the disease, corn leaf blight.

Plants of variety **A** were crossed with another inbred variety of maize, **B**, which had entirely green leaves and low resistance to the fungus. All the F_1 generation had entirely green leaves and low resistance.

- (a)** Using appropriate symbols and indicating what they represent, state the genotype of **A**.

..... [1]

The above F_1 generation was back-crossed to variety **A**.

- (b)** Draw a genetic diagram to explain the expected results of the F_2 generation from this cross.

[5]

[Total: 6]

4 Dengue fever is an infectious disease transmitted by a vector.

(a) (i) State the name of the pathogen that causes dengue fever.

..... [1]

(ii) State the name of the vector that transmits the pathogen.

..... [1]

In 2002, the World Health Organization estimates that there might be 50 million cases of dengue infection worldwide every year. In Singapore, dengue fever is endemic with year-round transmission. Fig 4.1 shows the incidences of dengue fever from 1985 to 3 September 2005.

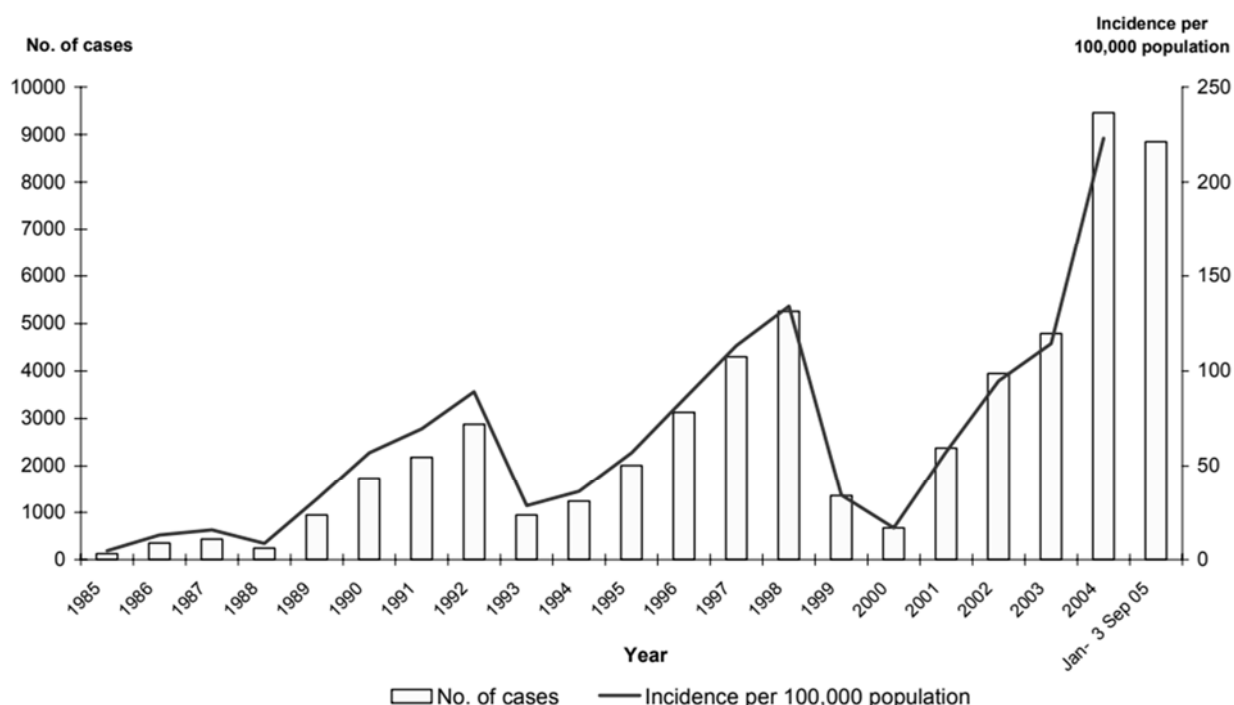


Fig. 4.1

(b) Suggest **two** reasons why the number of cases of dengue fever decreased in 1993 and 2000.

.....

.....

.....

..... [2]

- (c) Dengue fever is very difficult to control even though there has been improved understanding of the disease.

Explain why dengue fever is very difficult to control.

.....
..... [1]

[Total: 5]

- 5 Fig. 5.1 shows the Calvin cycle which takes place inside a plant cell.

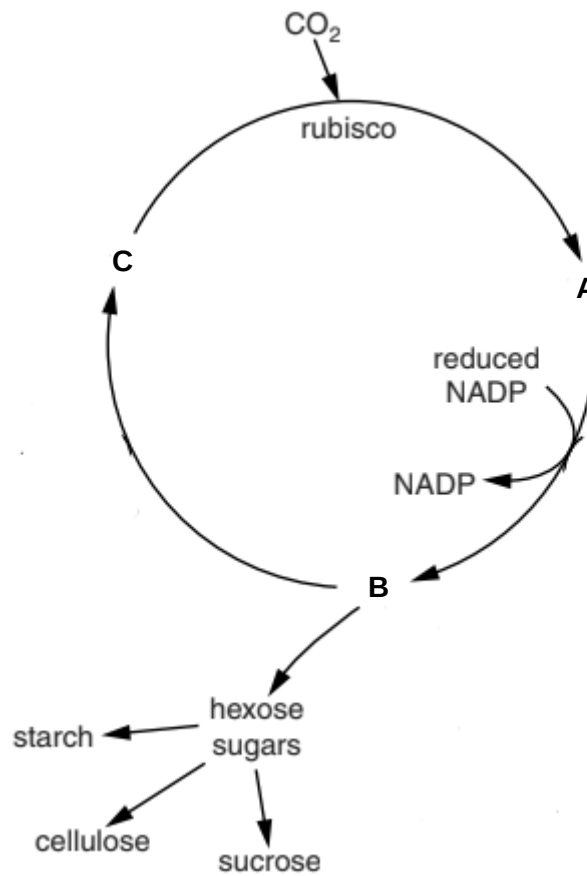


Fig. 5.1

- (a) Identify substances A, B and C.

A

B

C [3]

- (b) Describe the role of NADP in linking the light dependent reactions to the Calvin cycle in shown in Fig. 5.1.

.....

.....

.....

..... [2]

(c) In aerobic respiration and photosynthesis, ATP is synthesised via chemiosmosis in oxidative phosphorylation and photophosphorylation.

(i) Outline the role of stalked particles in chemiosmosis in oxidative phosphorylation.

.....

.....

.....

.....

.....

..... [2]

(ii) Explain why less ATP is synthesised from the same mass of glucose in anaerobic respiration compared to aerobic respiration.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

Write your answers on the lined papers provided at the end of this Question Paper.

Your answers must be in continuous prose, where applicable.

Your answers must be set out in sections **(a)** and **(b)**, as indicated in the question.

- Explain how haemoglobin is synthesised when haematopoietic stem cells form red blood cells. [9]

[Total: 15]

- Cytochrome c amino acid sequence in humans is more similar to that of chimpanzees but differs more from that of horses.

Explain how cytochrome c was conserved over time in eukaryotes. [9]

[Total: 15]

